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SURVEY

Generative AI in Engineering and Computing Education: A Scoping Review of Empirical Studies and Educational Practices

JONATHAN ÁLVAREZ ARIZA[1,2], (Member, IEEE), MILENA BENITEZ RESTREPO[2,3], AND CAROLA HERNÁNDEZ HERNÁNDEZ[2]

[1]Department of Technology in Electronics, Engineering Faculty, Corporación Universitaria Minuto de Dios-UNIMINUTO, Bogotá 111021, Colombia
[2]Engineering Faculty, Universidad de Los Andes, Bogotá 111041, Colombia
[3]Psychology Faculty, Universidad del Valle, Palmira 763531, Colombia

Corresponding author: Carola Hernández Hernández (c-hernan@uniandes.edu.co)

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ABSTRACT Since the release of diverse generative AI (GenAI) tools such as ChatGPT, Google Gemini, DALL·E, and GitHub Copilot, there has been much debate around the impacts and implications of these tools on education. Currently, extant literature remarks on the affordances, challenges, and opportunities of GenAI, but few studies report and analyze empirical studies and educational practices coming up by GenAI usage in learning settings. Then, in this Scoping Review (ScR) based on 146 studies retrieved from the databases SCOPUS, Web of Science (WoS), and ERIC, we analyzed the implications of integrating GenAI in engineering and computing education from K-12 to tertiary levels. We adopted an approach starting from the bibliometric features of the studies in terms of authors, cites, years, or cluster topics, and navigating to the identification of methodologies, strategies, AI literacy instruments and guidelines, learning outcomes, and students' and teachers' perceptions, among other features. We advocate that current educational practices in engineering and computing with GenAI can indicate to us a roadmap of its potentialities, uses, and risks from the standpoint of both teachers and students, and this could help us to create more reflexive methodologies that enhance the teaching-learning process based on the evidence. Our purpose with the outcomes and conclusions of this scoping review is to support educators, faculty members, and other stakeholders in engineering and computing education to co-create educational methodologies that articulate GenAI with curricula, AI literacy, and prompt engineering encompassing students' learning domains such as cognitive, affective, or behavioral.

INDEX TERMS Generative AI, GenAI, GAI, artificial intelligence, AI, computer science education, engineering education, computing education, prompt engineering, AI literacy.

I. INTRODUCTION

Since the release of ChatGPT, a pre-trained text generation language model by OpenAI on November 30, 2022 [1], much debate has emerged surrounding the implications and impacts of these kinds of generative AI (GenAI) tools could bring into education. Quickly, other big tech companies such as Google or Microsoft have joined in creating their own GenAI products such as Gemini, GitHub Copilot, Visual Studio, Open AI Codex, or Azure AI. As pointed out by Chan et al. [2], the term ''generative'' refers to the capability of AI to produce novel outputs rather than simply reproducing, categorizing, processing, and analyzing inputs. These outputs can come from several sources, including not only text but audio, video, and images, or in the case of engineering and computer science, programming code or design assistance [2], [3]. Similar definitions emphasize the capability of GenAI to

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create new and unique content, starting from pre-trained data [1].

At the technical level, GenAI works based on Large Language Models (LLMs) such as Generative Pre-trained Transformer (GPT-4) or (GPT-5), Gemini, Claude, Or LLa-Ma 2 [1], [4]. These models are sophisticated AI systems that can process and generate human-like language responses, working through *prompt engineering*. The idea behind prompt engineering is to enable computers to interpret and understand human language [1] and it has a close relationship with Natural Language Processing (NLP), which is an Artificial Intelligence field that examines how computer systems can interpret and control the natural language such as text or speech [5].

The prompts could be composed of four components, viz., context, instructions, input of content, and format [1]. The *context* refers to the role or person that AI takes on when it generates the responses. This creates more accuracy and contextualized responses. The *instructions* tell the IA what we want it to do. Therefore, the prompt requires clear instructions for the AI. The instructions can be used for diverse purposes, e.g., code documentation, solving code problems, bug reports, summarizing research papers, extracting key points in long emails, or identifying changes in a new software version library [1]. The *input of content* refers to the symbols like ### or """", which are boundaries that indicate to the LLM what information should be worked on. Finally, the *format* indicates to the LLM what the output format should be [1].

Complementary, Eager and Brunton [6] have defined other prompt components as being more orientated towards education. In this study, the prompts are composed of a *verb* that indicates a specific action to be performed, a *focus* providing the action to be performed, a *context* which is the scope or parameters for the task, an *alignment* that instructs AI to align content with the desired goals, and *constraints and limitations*. Using these prompt components, students can perform and participate in role plays, debates, or collaborative group projects [6]. For teachers, the prompts can help orientate question generation, rubric generation, lesson plans, or assessment [6].

These features of GenAI have drawn the attention of Educators and foremost Higher Education Institutions (HEIs), apparently revolutionizing the way to teach and learn and providing more educational tools to the students to complement their formation [4]. Among the advantages of GenAI for education are highlighted instant feedback, personalized learning, multilingual support, explaining complex concepts, and enhanced student engagement [4], [7]. GenAI systems, especially, in HEIs, have been encouraged by what is called digital transformation (DT) [8]. Quy et al. [8] define DT as integrating digital technologies into all sectors of an organization, leveraging technologies to change how it operates fundamentally and its business model. Among the benefits of HEIs are the prediction, planning, and control of the organization structure and human resources; improving the capacity and quality of training to create a good workforce; and changing of traditional management to an information technology-based one that relies on digitalized databases [8].

So, in the case of engineering education, GenAI tools have the potential to guide students in courses, e.g., of embedded systems, design, or programming [9] but keeping in mind the necessity that students evaluate critically the output generated that is aligned to the concept known as *AI literacy*, which we will address later on. Yet, GenAI technologies are premature with an initial state of development, and it is soon to grasp their effects in education [10], but certainly, we need to coexist effectively with them.

These advantages of GenAI are often accompanied by descriptors such as lower operating expenses, faster and more accurate decision-making processes, or increased productivity. Then, AI appears as a driver for labor market transformation, impacting job creation [11]. For instance, the Future of Jobs Report 2023 by the World Economic Forum (WEF) [11] puts forward that 75% of surveyed companies planned to adopt AI. At the same time, 50% of organizations expect to create job growth, and 25% expect to create job losses through its incorporation. The same report asserts that employers estimate that 44% of workers' skills will be disrupted in the next five years [11]. The issues related to this scenario, especially regarding job losses or layoffs have been among the most significant concerns about AI incorporation, and this is not unrelated to the educational institutions, i.e., HEIs. Employment growth is driven by the industrial sector, in which computer science and engineering play a crucial role, according to the Artificial Intelligence Index Report 2024 from Stanford University [12].

In this way, it seems that GenAI has come to stay, as several researchers have stated [13], [14]. Educators particularly for this review, Engineering and Computing Educators have much to do with this. Firstly, in leading students in proper usage and comprehension of the implications of AI technologies and secondly, in the creation of methodologies that help to improve. e.g., learning, engagement, and developing technical and soft skills such as communication, collaboration, and teamwork, integrating these cutting-edge GenAI technologies. Beyond the innovation obsession to use GenAI in educational institutions and their promoters or detractors, we have noticed a gap in the current literature in describing how engineering and computing Educators are integrating GenAI in their courses, or what educational outcomes students have developed. In addition, we observed a plethora of student-centered proposals during the review, but not teacher-centered ones that show their perceptions or initiatives, except, i.e., in references [15], [16], [17], [18].

Thus, in this scoping review based on PRISMA-ScR guidelines [19], we explore how GenAI is employed by engineering and computing Educators, and how they are leveraging these technologies' features to foster learning and engagement in engineering and computer science courses. Our objectives with this scoping review are the

following: (1). Identify the overall features of the studies in terms of bibliometric trends, such as cited articles, articles per year, leading countries, journals, authors, and cluster topics. (2). Inform researchers and/or Educators about the educational features of the studies identified, including features such as type of intervention, participants, research approach, or main learning outcomes. (3) Expose educational strategies elicited by the researchers utilizing GenAI. And (4). Expose issues, challenges and improvement opportunities for GenAI in engineering and computing education based on the evidence. We expect that this review can help researchers and Educators to clarify and identify the potentialities and constraints of GenAI for students and Educators in engineering and computing. We also encourage Educators to create their evidence-based educational methodologies, tailoring GenAI tools based on the results presented in this review.

II. BACKGROUND AND MOTIVATION

A. RELATED WORKS

We detected several related literature reviews and experiential studies concerning the scope of engineering, computing, and education, indicating mainly GenAI's affordances, challenges, and opportunities.

Crompton and Burke [7] identified the affordances of ChatGPT for education based on 44 studies. The affordances for educators include task automation and professional development, allowing the creation of effective instructional materials. For students, the affordances comprise explaining complex concepts, providing personalized feedback, and facilitating engagement.

Andrade-Girón et al. [20] examined the current scholarly landscape concerning the use of ChatGPT within higher education. The review emphasizes the positive impact of AI tools such as ChatGPT on student learning outcomes and time management. However, a significant research gap in rigorous experimental studies is identified, underscoring the need for more robust evidence through controlled settings.

Chen and Chen [21] evaluated the impact of AI on Education. The authors describe a set of categories in their review for AI in education as follows. Adoption and implementation, indicating that AI has been adopted through embedded systems, chatbots, and web-based, and online intelligent education systems. Customization and Personalization, which pose the idea that AI meets the individual needs and capabilities of students. Barrier Elimination assures that AI enables global access to education through online and web platforms. Instructional Effectiveness points out that AI overcomes the challenges of one-on-one teacher-student tutoring. At last, Academic Integrity in which AI has allowed integration of tools such as plagiarism checkers and online supervision tools, monitoring and ensuring the originality of students' work.

Robert and Mohamed [22] conducted a literature review based on 22 papers with SWOT analysis from 2022 onwards in computer science education. Concerning Strengths (S), Generative AI models, such as ChatGPT and Codex, demonstrate a solid ability to generate code effectively. Studies have shown promising results in solving fundamental programming problems. As for Weaknesses (W), GenAI models may exhibit issues related to reliability, leading to the generation of incorrect or inconsistent code without warning. As Opportunities (O) are highlighted helping beginner programmers, improvement of educator's productivity and workload, personalized learning, accessibility, and error feedback. Finally, as Threats (T), the authors indicate plagiarism and over-reliance on generated information through GenAI rather than on learned knowledge or other information sources. The study concludes that it is essential to recognize the capabilities of GenAI beyond academic misconduct.

Bahroun et al. [23] synthesized key insights of GenAI in educational settings based on selecting 207 research papers through a bibliometric analysis. The study points out several meaningful implications for Educators, researchers, and policymakers. For Educators, GenAI could bring pedagogical innovation, creating interdisciplinary collaboration, but at the same time, Educators need to dive into the ethical implications of GenAI, e.g., in data privacy, biases, and academic integrity. The study also encourages researchers to explore ways to make GenAI more accessible for diverse learners. Future studies should be focused on transparency, addressing biases, or designing AI-enhanced curricula for researchers. For policymakers, a regulatory framework for GenAI in education is necessary.

Regarding this latter part with regulatory frameworks or pre-emptive policies, Ifenthaler et al. [24] presents a Delphi study based on a survey administered to 33 international professionals, indicating as outstanding results three trends: (1) privacy and ethical use of AI, (2) the importance of trustworthy algorithms, and (3) equity and fairness. The study highlights the importance of precautionary pre-emptive policies preceding practice for AI in education. Decision-makers may introduce such policies in response to societal concerns, rapid technological advancements, and the need to align educational practices with ethical and legal standards.

Lodge et al. [25] have identified several critical areas of tertiary education that will be affected by GenAI. Firstly, Educators are encouraged to rethink the role of the assessments transforming the traditional ones and maintaining academic integrity after using AI's capabilities. Secondly, it is necessary to prepare students to cope effectively with AI technologies in the job market by analyzing the social and ethical implications of GenAI. Also, the authors claim there is a need for an agenda to research the capabilities and applications of GenAI in education.

By the same token, Pather et al. [26] discuss GenAI in computing education through a study in two stages. The first one identifies relevant studies in computing education and GenAI and describes them in a literature review. The second one exposes the findings of a survey showing the prevailing perceptions of teachers and students across 20 countries in

the field of computing education and GenAI. The authors reported that students and teachers have limited experience with AI tools. Moreover, some risks and concerns were identified such as academic integrity, ethical use of AI tools, over-reliance, and the need for policy development and the clarification about how GenAI could be used in the coursework.

Finally, Santos et al. [27] describe a literature review on GenAI within Information and Communication Technologies (ICT) engineering education. The study shows the potential of GenAI technologies to enhance students' educational process. In addition, the study explores the impact of GenAI on pedagogical strategies, curriculum development, and student engagement in engineering education. The study entails that it is necessary to adapt curricula to encompass unique skills in AI literacy.

B. AI LITERACY

AI literacy is the ability to comprehend, assess, interact with, and make informed decisions using AI technologies [2]. AI literacy involves understanding the basic principles of AI, recognizing its applications, and being aware of the ethical, social, and privacy implications of the system using it [2]. Long and Magerko [28] have defined AI literacy as a set of competencies enabling individuals to critically evaluate AI technologies, and communicate and collaborate effectively with them. Herein, Digital Literacy is a prerequisite for AI literacy since it is necessary to know how to use computers to make sense of AI. The concept of AI literacy is helpful if we analyze the limitations and pitfalls of GenAI reported by educational researchers, which include plagiarism and cheating, privacy issues, potential bias, diversity issues, or spread of false information [7]. Rather than follow-up disruptive technologies such as GenAI, it is better to develop skills in the students and teachers to evaluate and understand the affordances and limitations of thereof [10], [29].

It is common to find studies evinced the features for assessment and feedback in courses of CS1 or CS2 or establishing the potentialities of GenAI for them, typically utilizing GPT-3 or GPT-4, while at the same time the authors mention potential risks or misbehaviors, which comprise cheating or plagiarism practices by the students [30], [31], [32]. Nonetheless, Lee et al. [33] propose that at least in high schools, GenAI technologies such as ChatGPT have not proven to increase cheating practices. Indeed, cheating practices were high before the release of ChatGPT and might have been exacerbated by the context of the COVID-19 pandemic. However, new studies in other educational settings are needed to endorse or refute these results [33]. Haugsbakenm and Hagelia [34] have discussed the idea that it is better to talk about the term of Educational *Promptization* instead of prompt engineering because the current literature depicts a big concern about the efficiency of the prompts to generate information. The concept of *promptization* implies that apart from designing precise and effective prompts, students and educators can assess and interpret their outputs, which involves critical thinking and reflection on the information retrieved. Even prompts can scaffold students' learning process, guiding them step-by-step in different tasks [34]. However, the interaction with GenAI chatbots depends on prompts, which can influence the acceptance of the information generated. Students who are less familiar with prompts or have experienced negative experiences with GenAI chatbots prefer not to use them [35]. Similarly, students who consider AI models as black boxes do not trust them [36].

Searching to define a framework for AI literacy, Long and Magerko [28] have defined 16 competencies that encompass actions from learners such as recognizing AI technological artifacts, distinguishing between narrow and general AI, encourage learners to create training datasets, identify key ethical issues of AI, critically interpret data, understand the steps involved in AI and machine learning models, among others. In parallel, the authors provide a set of design considerations for educators using or developing AI systems. These considerations describe actions such as explainability, contextualizing data, promoting transparency and critical thinking, considering students' identity, values, and backgrounds, or leveraging students' interests. Another framework proposed for GenAI in education is CRISPE [37], [38] which has the following stages:

- Capacity and Role (**CR**): What capacity or role will be adopted by GenAI like ChatGPT? The role could be a researcher, educator, student, etc.
- Insight (**I**): Provide any contextual information or background in detail.
- Statement (**S**): What should GenAI do? This description should be precise, entailing as many details as possible.
- Personality (**P**): What would be the tone for the output produced by GenAI: Formal, informal, business-oriented, education-oriented, etc.?
- Experiment (**E**): Ask GenAI multiple times, choosing the better answer according to the user's needs.

Educators or researchers interested in other frameworks for AI literacy can consult the ones proposed by Kong and Zang [39] which involve four dimensions: cognitive (understanding AI concepts), affective (psychological readiness to use AI), metacognitive (AI for problem-solving), and social (Ethics and problem-solving); and Kandlhofer et al. [40], which entails topics of AI taught to the students from kindergarten to university. Both frameworks were tested with students, exploring the connection of basic AI concepts with applications in daily life.

C. MOTIVATION

As educators, we have observed how GenAI has been widely promoted as a new tool seemingly revolutionizing the educational landscape and how students learn, making them more efficient, productive, and engaged in diverse tasks in education settings. At the same time, we have noticed that research on GenAI in education is focused on

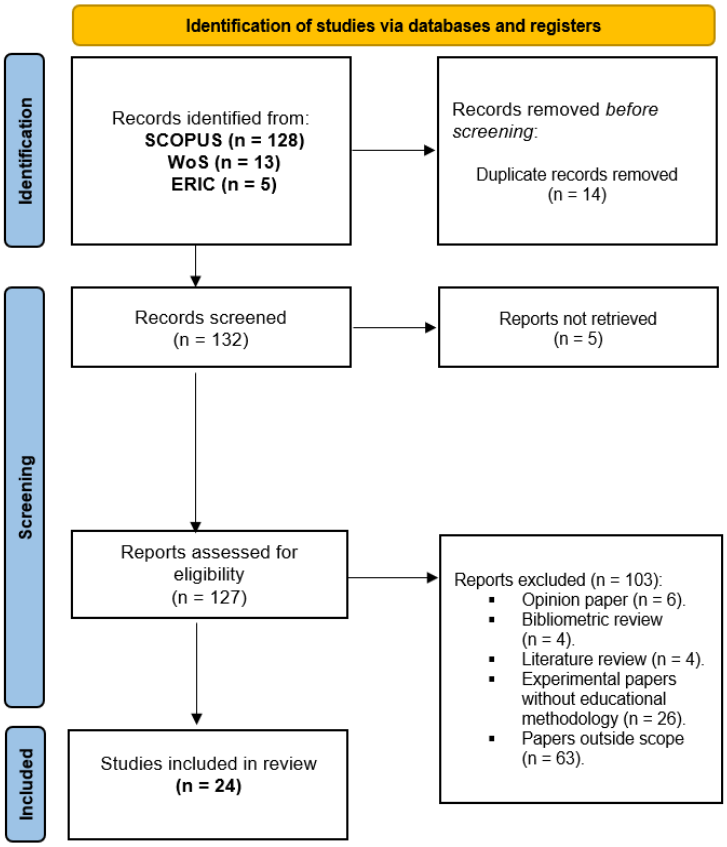


FIGURE 1. PRISMA-ScR flow diagram tailored for the study.

its affordances and potentialities instead of discussing its implications from a pedagogical standpoint based on the evidence. We think that the interaction between GenAI tools, teachers, and students can lead to a better understanding of these and the design of frameworks or methodologies that can incorporate them. Then, this study seeks to address the lack of studies pedagogically oriented, analyzing and describing how GenAI is employed or tailored by Educators and students, and what learning outcomes, difficulties, standpoints, strategies, or limitations have emerged from its use. We utter that it is the mediation and reflection that involves practices among students, Educators, and the technological tools, in this case, GenAI, that can bring out good results and transformation in education.

III. METHOD

We performed a scoping review based on PRISMA-ScR guidelines [19]. Since the field of the review (GenAI) is relatively new, we performed a scoping review instead of a systematic literature review (SLR). In addition, the nature of scoping reviews is exploratory aiming at rapidly mapping the key concepts in a research area [41]. Scoping reviews summarize and disseminate research findings, identifying research gaps in the existing literature, or examining the range and nature of research activity [41]. We tailored for the study the stages for scoping reviews proposed by Arksey and O'Malley [41], which are the following: Identifying the research question; identifying relevant studies; study selection; charting the data; and collating, summarizing, and reporting the results. Fig.1 depicts the flow diagram adopted through PRISMA-ScR guidelines for this study.

A. RESEARCH QUESTIONS

To meet the objectives of this study, we proposed a set of research questions (RQs). Table (1) depicts these questions with their respective descriptions.

For the study, we adopted an approach starting from the general features of the studies (bibliometric trends and research topics) and navigating towards their pedagogical particularities that help understand how Educators utilize GenAI in engineering and computing education from K-12 to tertiary levels. It is worth mentioning that this scoping review focuses on empirical research with evidence of application and evaluation whether or not the final educational results were positive or negative. Indeed, our intention is based on the evidence to describe the challenges and recommendations of using GenAI as RQ4 indicates.

B. IDENTIFICATION OF STUDIES

To identify relevant studies, we searched them in three databases: SCOPUS, Web of Science (WoS), and Education Resources Information Center (ERIC) with the respective

TABLE 1. Description and purpose of the RQs in the scoping review.

Research Question (RQ)	Description and purpose
RQ1. What are the bibliometric characteristics of the studies in terms of citations, journals, authors, and research topics?	Identify relevant studies, authors, journals, and research topics to illustrate how GenAI is employed in engineering and computing education.
RQ2. What are the educational characteristics of the selected studies and their outcomes in the cognitive, affective, or behavioral learning domains?	Describe the educational features of the selected studies altogether with their learning outcomes in the cognitive, affective, or behavioral learning domains.
RQ3. What Gen IA integration strategies and activities have Educators and/or researchers constructed?	Explain the strategies and activities developed by Educators or Researchers to integrate GenAI in engineering and computing education.
RQ4. What are the challenges, issues, and opportunities for improvement of GenAI in engineering and computing education?	Describe the issues, challenges and opportunities for improvement based on the studies analyzed.

TABLE 2. Description of searching criteria for the scoping review.

Aspect	Description
Databases	SCOPUS, Web of Science (WoS), ERIC
Searching Strings	• **SCOPUS**: TITLE-ABS-KEY((generative AI) AND ("engineering education" OR "computing education")) • **WoS**: TS=((generative AI) AND ("engineering education" OR "computing education")) • **ERIC**: (generative AI) AND ("engineering education" OR "computing education")
Inclusion Criteria	• Empirical research integrating or developing GenAI in engineering and computing education with evidence of application in learning settings. • Language: English. • Type of study: Primary research (Data collected directly for the researchers in classrooms or learning settings).
Exclusion Criteria	Studies outside the scope of the review: not GenAI integration in engineering and computing education, not evidence of application and learning outcomes developed, not in English, not primary research, not preprints.
Total records obtained	SCOPUS (128), WoS (13), ERIC (5). **Total Records**: 146.

string queries, inclusion, and exclusion criteria reported in Table 2. Fig.1 illustrates the number of registers collected in each database. Specifically, we included the terms ''engineering education'' and ''computing education'' with the tags in each database to search directly in the title, abstract, or keywords of every record obtained with the boolean connectors **AND** and **OR**. Regarding generative AI, we prefer not to include abbreviations such as GenAI or GAI since studies typically include the complete term, even in their indexing terms. Also, we do not add a specific timeframe because GenAI is relatively a new phenomenon. However, the studies retrieved are between 2022 and 2024. We obtained 146 records with this procedure.

C. SCREENING AND APPRAISAL CRITERIA

We downloaded the records of each database ($n = 146$) in the format of research information system (RIS). Then, these records were added to the software Rayyan.ai [42], [43], which is a software specialized in the screening process for systematic and scoping reviews. With this software, firstly, we checked duplicates ($n = 14$), and secondly, we screened the ($n = 132$) remaining records. Nonetheless, ($n = 5$) articles could not be retrieved because the missing DOI or publisher was unavailable. Finally, ($n = 127$) records were retrieved.

In the latter studies, we searched for integration of GenAI in the educational settings in engineering and computing through an approach such as quantitative, qualitative, or mixed in any educational level (K-12 to tertiary). In addition, the studies must exhibit the development of any of the learning outcomes in the levels of cognitive (knowledge and skills), affective (attitudes, motivation), or behavioral (developed behaviors, e.g., socialization, communication, and organization skills), which are called domains of learning [44], [45].

We noticed that plenty of the studies were opinion papers, articles exposing challenges and opportunities of GenAI, experience papers, or experimental papers often reporting benchmarking about how ChatGPT can pass programming exams in CS1 or CS2 courses or describing possible behaviors of the students with GenAI usage such as cheating or plagiarism. Only a few studies reported methodologies incorporating GenAI for educational purposes with evidence of any approach mentioned.

The screening process started with reading the title and abstract of each of the 127 records under the lens of the including and excluding criteria described in Table 2. Independently, three coders performed the previous procedure, initially getting a Cohen's Kappa coefficient ($\kappa = 0.84$). Cohen's Kappa coefficient measures the degree of agreement between raters in which the values of 0.4 to 0.6 are fair, 0.6 to 0.75 good, and over 0.75 excellent [46]. When we found a controversial article, that is, an article that could be approved

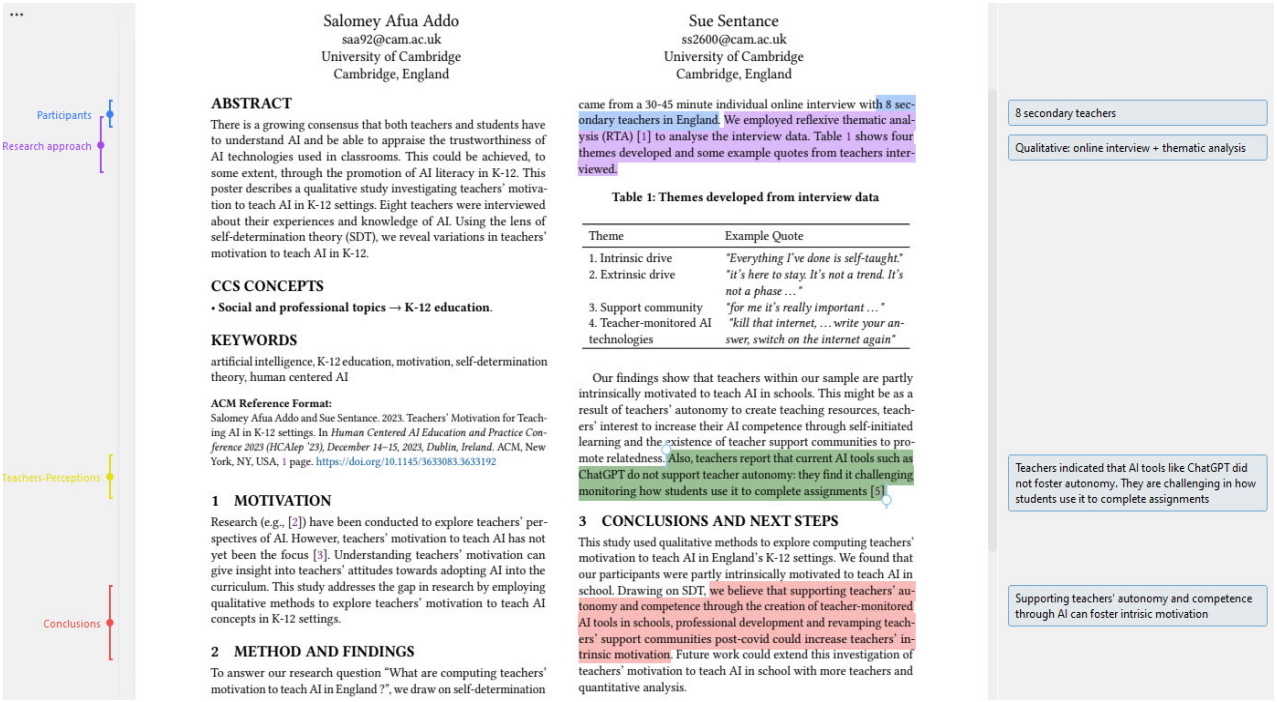


FIGURE 2. Example of information coded using MAXQDA for each study.

but we hesitated, altogether we read the whole article and approved or excluded it, finally, achieving 100% agreement.

As our interest was focused on empirical research, we classified each article according to the framework developed by Montesi and Lago [47] in the following categories:

- Case studies
- Experimental studies
- Meta-analyses
- Opinion studies
- Experience studies
- Literature reviews

Under this scheme, we identified ($n = 24$) studies that comply with the criteria of Table 2 with 100% of agreement between the coders. These articles fall into the categories of case and experimental studies.

With these studies, to respond to RQ1, VOSViewer v.1.6.14 [48] and Crossref REST API [49] were employed. VOSViewer is a software for bibliometric analysis based on network data that is focused on items and clusters with two overall functions: creating maps and visualizing them. VOSViewer was focused on the keywords of every study approved. Crossref REST API is a tool that allows the extraction of bibliometric information such as citations, authors, publishers, type of study (journal article, conference proceedings, book chapter), or articles per year, among others. The API was used with Python language to extract the bibliometric features of the studies.

To respond to RQ2, we extracted information from each approved article such as study aim, research approach, participants, educational level, learning outcomes, and conclusions. To respond to RQ3, we identified some strategies and curricular activities described by the researchers in the learning settings. The information for RQ2 and RQ3 was coded using MAXQDA 24 [50]. We adopted the following procedure. Two researchers (A and B) coded 640 fragments for the 24 studies, reading the articles and extracting the information. The articles were divided into two groups of $n = 12$ studies, and each group was assigned and analyzed independently by two researchers who served as coders. Next, jointly, each code was revisited and discussed by the coders (A, B), creating a matrix with information such as research approach, participants, conclusions, AI instruments and guidelines, and students' and teachers' perceptions, among other features. Under this procedure, we adopted an open coding approach [51], recognizing the features of each study and identifying emergent topics. So, we reached an inter-coder agreement (ICA) of 100%.

Fig.2 depicts an example of information coded. We also included several comments for each article to identify, e.g., educational strategies, researchers' insights, or any additional information crucial for the review. Fig.3 shows an example of the codes created in the process.

Similarly, to respond to RQ4, we analyzed the conclusions indicated by the authors in the studies and we identified the challenges, recommendations, and improvement opportunities.

D. REPORTING THE RESULTS

Guidelines proposed by Webster and Watson [52] were considered for writing the review in aspects such as



FIGURE 3. Example of codes created in MAXQDA for the approved articles.

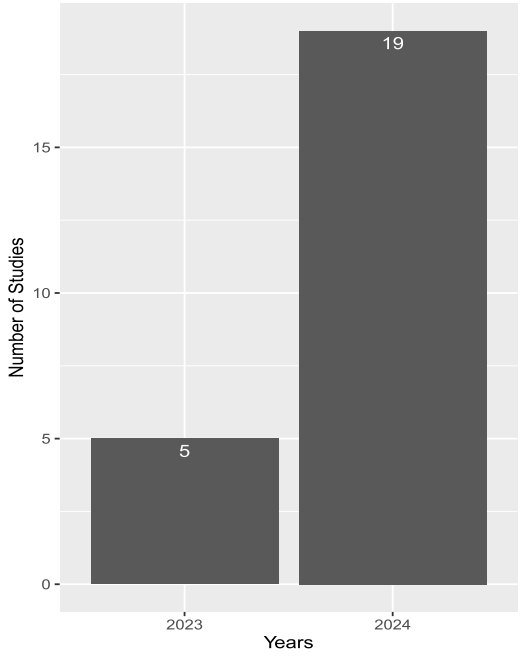


FIGURE 4. Number of studies by year. N = 24.

identifying relevant literature, review based on a concept-centric approach, tables and figures presentation, tone and structure of the synthesis. Results are presented according to the described Research Questions (RQs).

IV. FINDINGS AND DISCUSSION

A. RQ1. WHAT ARE THE BIBLIOMETRIC CHARACTERISTICS OF THE STUDIES IN TERMS OF CITATIONS, JOURNALS, AUTHORS, AND RESEARCH TOPICS?

1) YEARS

Fig.4 depicts the number of studies per year. The trend shows an increase in proposals from 2023 to 2024 consistent with other prior systematic reviews [53], [54]. None proposals were identified for 2022.

In 2023, the most cited article *Unleashing ChatGPT's Power: A Case Study on Optimizing Information Retrieval in Flipped Classrooms via Prompt Engineering* [38] describes a GenAI methodology with ChatGPT for computer science students, specifically, for the topic of TCP socket programming. Using the CRISPE framework [37], students could create more comprehensive and complete prompts to get in-depth information about the topics addressed. Another trend for this year is the concentration of proposals addressed from the teachers' perspective (see studies [15], [18], [55]). These studies show the teachers' perceptions regarding GenAI, their possibilities, and the concerns related to their deployment in education.

In 2024, the most cited article *Prompt Problems: A New Programming Exercise for the Generative AI Era* [56] deploys a GenAI methodology for CS1 and CS2. The methodology only offered visual representations of the problems, and students should use GenAI (ChatGPT) as a guide to achieve a solution. Apart from the methodology, the authors created an authoritative prompt tool called *Promptly*. It is interesting to observe that by 2024, several studies have proposed their own prompting tools with particular educational features for the courses [56], [57], [58]. In addition, proposals with GenAI for children as co-creators of digital media with music, storytelling, and arts were identified in this year (see studies [59], [67]). The authors employed Dall-E, ChatGPT, and Runway ML as GenAI tools in these studies.

Also, for 2024, we noticed that the discussion moved toward other educational insights of GenAI apart from the perceptions of students and teachers. For instance, Okolo [66] describes a set of curricular activities with GenAI to teach computing to high school students, while Hou et al. [35] explain the help-seeking preferences of CS students with ChatGPT, Midjourney, or GitHub Copilot. It turns out that Educators in engineering and computing are turning their sights to other concerns about GenAI, including pedagogical ones, even though with a limited number of proposals.

2) COUNTRIES

Of 24 studies analyzed, 11 proposals came from the U.S (45.83%), two proposals from the U.K. (8.33%), two proposals from Guatemala (8.33%), two proposals from Germany (8.33%), and countries such as New Zealand, Spain, Chile, Sweden, China, Finland, and Serbia each one with a proposal (4.16%) respectively. In the case of Guatemala, both proposals entail the teachers' perceptions regarding GenAI in computing and engineering education. Then, mainly, GenAI educational research has been led by the U.S.

3) TOP-CITED ARTICLES, AUTHORS, AND PUBLICATION TYPE

Table3 shows the top-cited articles in the scoping review, with the authors and citations. Herein, the most cited article by Wang et al. [38] and published in the journal IEEE Transactions on Learning Technologies, describes a flipped classroom methodology supported with prompt engineering using ChatGPT for 26 undergraduate students. For this study,

TABLE 3. Top-cited articles and authors in the Scoping Review. Ref. Study reference.

#	Authors (Year)	Study Title	Citations	Ref.
1.	Wang et al. (2024)	Unleashing ChatGPT's Power: A Case Study on Optimizing Information Retrieval in Flipped Classrooms via Prompt Engineering	21	[38]
2.	Denny et al. (2024)	Prompt Problems: A New Programming Exercise for the Generative AI Era	16	[56]
3.	Liu et al. (2024)	Teaching CS50 with AI: Leveraging Generative Artificial Intelligence in Computer Science Education	12	[57]
4.	Hou et al. (2024)	The Effects of Generative AI on Computing Students' Help-Seeking Preferences	10	[35]
5.	Sheard et al. (2024)	Instructor Perceptions of AI Code Generation Tools - A Multi-Institutional Interview Study	4	[17]

the authors used the framework CRISPE to create effective prompts.

In the second place, Denny et al. [56] report the experience of CS1 and CS2 students in formulating proper prompts to solve CS1 and CS2 programming problems through visual representations. The students should use these representations and utilize ChatGPT as an assistant to reach the problems' solutions. In the third place, Liu et al. [57] show how to integrate GenAI into an introductory programming course with a significant enrollment rate. The authors highlight the importance of GenAI to approximate 1:1 teacher-to-student ratio in the course. In the fourth place, Hou et al. [35] analyze the help-seeking preferences of the students with GenAI. The study entails how ChatGPT and GitHub Copilot are comparable to human help, and even in some cases students prefer these tools as assistants for their learning. At last, Sheard et al. [17] discuss the instructors' perceptions regarding GenAI, and its possibilities and challenges for students and Educators.

Other studies such as [16], explore teachers' perceptions of GenAI counts with three cites, [60] describing the AI literacy and its relationship with prompt strategies with three cites, [61] showing how to integrate GenAI as an assistant for design in engineering with two cites, [18] analyzing the role of ChatGPT from the perspective of teachers, and [62] proposing educational strategies for computer science and engineering with one cite, complete Table 3.

As for publication type in the studies, ($n = 19$, 82.6%) are conference proceedings while ($n = 5$, 21.74%) are journal articles. Fig. 5 depicts the percentage of publishers for the studies. Most of the proposals were published in

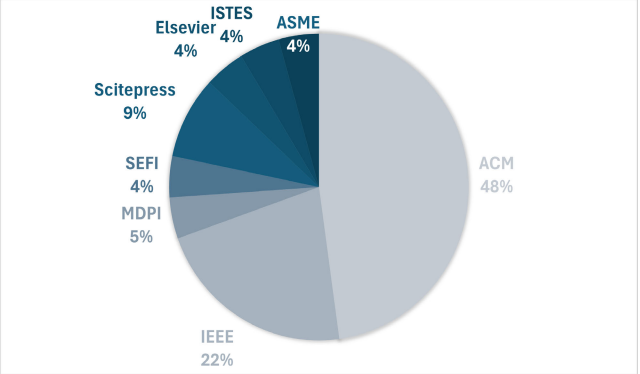


FIGURE 5. Distribution of the publishers for the N = 24 studies in the ScR. European Society for Engineering Education (SEFI), The American Society of Mechanical Engineers (ASME), International Society for Technology, Education, and Science (ISTES).

TABLE 4. List of the top-cited publication sources in the ScR. Ref. Study reference.

#	Publication Source	Citations	Ref.
1.	Proceedings of the 55th ACM Technical Symposium on Computer Science Education	34	[17], [56]–[59]
2.	IEEE Transactions on Learning Technologies	21	[38]
3.	Proceedings of the 26th Australasian Computing Education Conference	11	[35]
4.	2023 IEEE Frontiers in Education Conference (FIE)	4	[18], [62]
5.	Computers and Education: Artificial Intelligence	3	[60]

Association for Computing Machinery (ACM) conferences ($n = 11$, 48%) followed by IEEE ($n = 5$, 22%). Table 4 shows the titles of conferences and journals with more citations in the ScR. This manifests the trend that the central part of the studies is related to computing education and computer science ($n = 17$); meanwhile, several proposals ($n = 7$) are for engineering education.

4) RESEARCH TOPICS

Fig.6 illustrates the cluster map for the keywords in the study. There are some noticeable aspects to remark. Firstly, AI literacy and AI pedagogy are related to arts, computing, and music for elementary school students (see dark blue cluster and studies [59], [67]). In the same cluster, AI literacy appears as a 21st-century skill to be developed by students. Also, for high school students, the study [66] proposes a set of activities to strengthen AI knowledge and literacy.

Secondly, the green cluster (Computing Education) is related to code generation, prompt engineering, conflict resolution, chatbots, and flipped classrooms. For instance, many studies for computing education in the ScR used

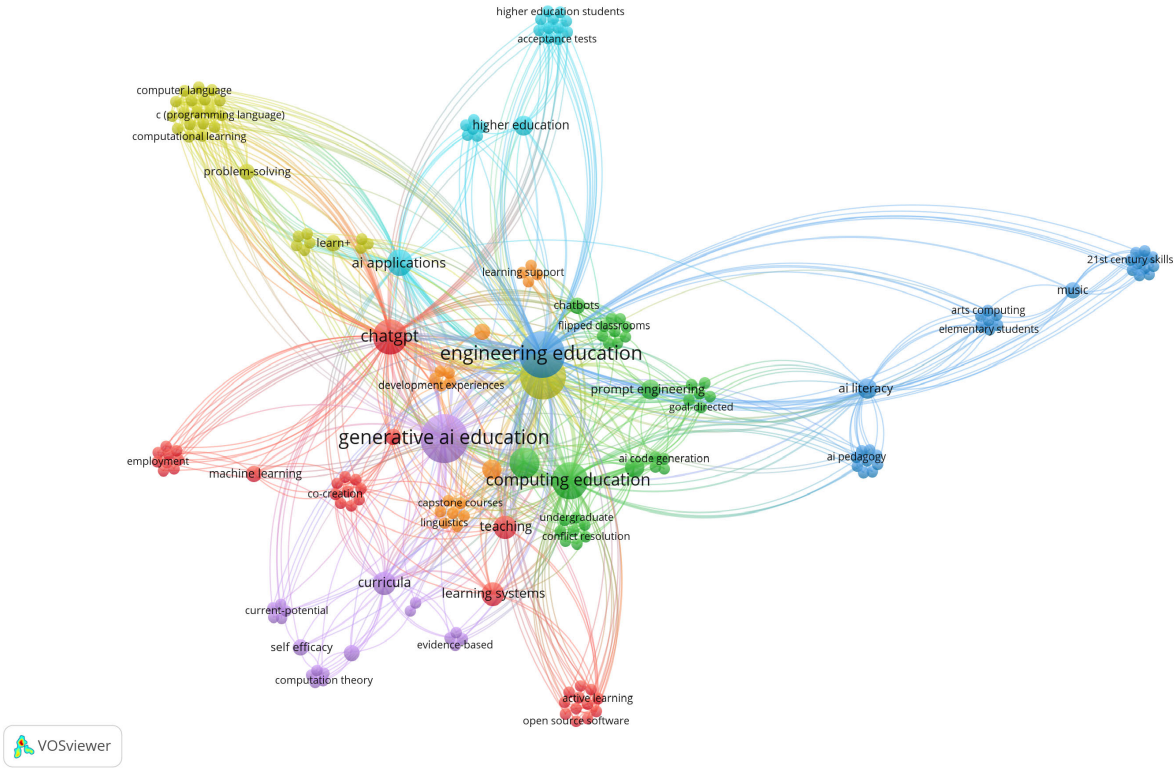


FIGURE 6. VosViewer cluster map for the keywords in the studies.

GenAI tools for code generation, making that students paid attention to learning the logical structure of their programs and contrasting the new knowledge acquired by using GenAI with the prior one [35], [56]. Conversely, for the same feature of GenAI, Choudhuri et al. [63] have reported problems, specifically, hallucinations and missing details and critical technical information in the case of ChatGPT. Seemingly, ChatGPT can work well with ''generic information'' for CS1 and CS2 courses, but for more specific topics such as TCP sockets or encryption, retrieved information can fail, even providing wrong guidance [63]. Another keyword in the same cluster is conflict resolution. So, there have been other uses of GenAI for computing and engineering education, which include conflict resolution (GenAI as an assistant to provide advice to solve conflicts) [64] and analysis of engineering students' CVs for entrepreneurial activity [65]. Also, the keyword prompt engineering has a relationship with AI literacy (see the closeness of these concepts in clusters of dark blue and green). Nevertheless, while it seems that a crucial factor in the learning of students with GenAI is their capability to formulate properly quality prompts to generate information, Knoth et al. [60] have suggested that not necessarily students need to know about AI literacy to create these prompts. This deserves more exploration in further studies because competencies of AI literacy are still broad, and these cannot be applied to all educational levels (K-12 to university) (see AI competencies formulated by Long and Magerko [28]).

Thirdly, ChatGPT has been the GenAI tool most utilized in the studies (see red cluster). Some features of active learning and engagement attributed to the tool have been fostering problem-solving skills and computational learning (see green and yellow clusters), enhancing curricular activities created by teachers, and providing learning support as an assistant, among others. Nonetheless, in the case of teachers, it is not clear the appropriateness and confidence levels that teachers have with GenAI tools [16], which can evince that teachers need more accompaniment from a pedagogical standpoint to orientate and employ GenAI in classrooms.

Finally, other keywords in small clusters (see purple, orange, and aquamarine ones) include self-efficacy that can influence teachers at the moment to create activities with GenAI [15], [55], higher education, and development of experiences with GenAI in capstone projects [58].

B. RQ2. WHAT ARE THE EDUCATIONAL CHARACTERISTICS OF THE SELECTED STUDIES AND THEIR OUTCOMES IN THE COGNITIVE, AFFECTIVE, OR BEHAVIORAL LEARNING DOMAINS?

Tables 5 to 10 depict the synthesis of the studies in the ScR. We classified this synthesis into student-centered proposals and teacher-centered ones. We also divided the synthesis into several educational levels from K-12 to university. Readers can notice the prevalence of student-centered computing education proposals focused on cognitive learning outcomes at the university level. Nonetheless, for elementary and

TABLE 5. Student-centered proposals for K-12 levels. Learning outcomes developed C: Cognitive, A: Affective, B: Behavioral.

#	Authors	Study aim	Research Approach	Participants	Outcomes-conclusions	Learning domains developed
1.	Garcia et al. [59]	Explore how elementary-aged Latinx students historically excluded in CS establish connections between community, heritage, and technology using a text-to-image Generative AI (DALL.E).	Qualitative (Focus group)	9 students, elementary school	Elementary students can make deeper connections when discussing imagined futures of their community and technology, creating prompts that embed technology into cultural practices.	A, B
2.	Okolo [66]	Engage underrepresented secondary school students in a 5-day workshop on how to use AI responsibly, understanding the socio-ethical implications of GenAI emerging technologies.	Mixed (pre and post-survey with open-ended questions)	4 students, middle school	Critically engaging students in lectures, reflecting at the same time on their usage of AI tools can help them to understand potential harms. It is needed to identify the initial AI knowledge of students.	C, A
3.	Parada-Cabaleiro [67]	Explore in a STEAM setting how AI literacy might be promoted in the music classroom through audiovisual co-creativity and digital storytelling.	Mixed (Survey + Observation)	9 students (10–12-year-old)	Students were aware of some potential risks related to AI. AI-child co-creativity within a digital storytelling framework might be a promising method to promote several lifelong learning competencies.	C, A, B

TABLE 6. Student-centered proposals for university level. Learning outcomes developed C: Cognitive, A: Affective, B: Behavioral.

#	Authors	Study aim	Research Approach	Participants	Outcomes-conclusions	Learning domains developed
1.	Garcia Huertes & Bragós Bardia [65]	Propose a novel methodology for evaluating entrepreneurial activity among engineering students. GenAI is used to infer entrepreneurial activity by analyzing Curriculum Vitae (CVs) of students.	Exploratory analysis	200 graduated engineering students	By using GenAI, Business Education promoted in undergraduate courses in Engineering is positively associated with entrepreneurial activity among engineering education alumni.	B
2.	Lauren & Watta [62]	Explore the innovative use of Generative AI using ChatGPT in computer science and engineering education to support learning outcomes as specified for each course.	Mixed (analysis of students' interactions with ChatGPT + students' comments analysis)	25 CS and mathematical students, 20 electrical and computer engineering students (undergraduate)	GenAI applications are useful in computer science and engineering education. The study evinced four strategies with GenAI: Well-defined learning goals, authentic learning experiences, structured learning progression, and strategic assessment.	C
3.	Denny et al. [56]	Understand how students might use LLM tools like ChatGPT, employing a prompt writing assignment with CS1 and CS2 problems. Also, researchers developed a web-based tool for GenAI called *Promptly*.	Quantitative (analysis of students' interactions with ChatGPT)	15 CS graduate students, 58 CS1 students, 182 CS2 students.	Some students reported hesitation about automatic code generation and others expressed anxiety about the feature of AI code-generating models. Future works should investigate the right time to introduce students to prompt-based code generation.	C
4.	Stankovski et al. [68]	Analyze the use of generative AI tools in engineering education, specifically in the subject of Programmable Logic Controllers (PLCs).	Quantitative (Survey + descriptive statistics)	41 industrial engineering students + 43 mechatronics undergraduate students.	Students obtained more correct answers using GenAI because of the experience gained in asking questions to ChatGPT and checking its answers. 97.37% of students hesitated about the answers given by ChatGPT.	C

TABLE 7. Table 6 continued. Student-centered proposals for university level. Learning outcomes developed C: Cognitive, A: Affective, B: Behavioral.

#	Authors	Study aim	Research Approach	Participants	Outcomes-conclusions	Learning domains developed
5.	Aggrawal & Magana [64]	Implement a conflict management training approach guided by principles of transformative learning by using LLMs. The approach involved learn, practice, and reflect stages, utilizing ChatGPT.	Qualitative (open coding approach + Analysis of ChatGPT conversations)	56 undergraduate engineering students.	Five conflict resolution styles were identified, and they were used by the students. The conflict resolution styles were Avoiding, Accommodating, Compromising, Competing, and Collaborating. 65% of the students increased their confidence in managing conflict and 63% of the students identified the problem's root (conflict origin).	C, B
6.	Choudhuri et al. [63]	Identify how effective is GenAI in helping students in software engineering tasks and establish the current pitfalls of GenAI. The authors propose a set of hypotheses to contrast through the evidence.	Quantitative (Quasi-experiment)	29 software engineering students (15 experimental group, 14 control group) (undergraduate)	Researchers create a small program to manage task complexity with GenAI. Frustration levels in students using ChatGPT were significantly higher vs. the control group. Moreover, researchers did not find an impact on productivity for students employing ChatGPT. ChatGPT struggles with specific computer science topics.	C
7.	Shaer & Cooper [61]	Include GenAI in the co-creation of activities in different stages of project development (design a novel tangible interface) such as ideation, conceptual design, and prototyping.	Qualitative (Analysis of students' experiences)	20 undergraduate engineering students	Results indicate that using GPT-3 during early ideation helped students to expand their ideas. However, GPT-3 lacks creativity and common sense. 25% of students reported problems with prompt engineering. Most students reported challenges working with DALL.E.	C, B
8.	Liu et al. [57]	Evaluate the usage and educational implications of non-commercial GenAI tools (CS50.ai) designed by the researchers in the study. The tools were deployed in a CS50 (introduction to programming) course.	Quantitative (Prompts' analysis + survey)	70 computer science students (initial stage). 500 on-campus students (second stage)	Providing students with own developed AI tools by teachers can reduce misbehavior such as cheating or plagiarism. The Anthropomorphic features of GenAI tools developed by the Authors contributed to their success. Most students found the course's AI tools beneficial.	C, B
9.	Wassem et al. [69]	Assessing ChatGPT's effectiveness, benefits, limitations, and its influence on learning in software development.	Quantitative (survey + descriptive statistics)	7 undergraduate software engineering students	Students perceived AI tools such as ChatGPT as beneficial, especially for complex projects. ChatGPT was generally perceived to positively impact various facets of software development learning and skill development.	C
10.	Hou et al. [35]	Investigate the help-seeking preferences and experiences of computing students now that generative AI tools are available to them.	Mixed (survey + semi-structured interviews + thematic analysis)	47 undergraduate computer science students	Students who were willing and able to iterate and reformulate their feedback requests described having more success. The quality of assistance received is intricately linked to the clarity and precision of the prompts.	C, A
11.	Ouhbi [70]	Implement active learning strategies within an active learning classroom setting and redesigned the project work using GenAI.	Qualitative (Analysis of students' comments)	54 undergraduate software engineering students	GenAI could be a useful tool to help students understand the code, identify code sections to be refactored, debug, write tests, and break dependencies.	C

TABLE 8. Table 6 continued. Student-centered proposals for university level. Learning outcomes developed C: Cognitive, A: Affective, B: Behavioral.

#	Authors	Study aim	Research Approach	Participants	Outcomes-conclusions	Learning domains developed
12.	Wang et al. [38]	Evaluate the ChatGPT effectiveness in task completion in a flipped classroom on TCP socket programming.	Quantitative (Quasi-experiment)	26 undergraduate computer science students (Control group: 13 students with no prior knowledge on the subject, 13 students with prior knowledge)	Prompts generated by the CRISPE (Capacity-Role; Insight; Statement; Personality; Experiment) framework can elicit more complete and in-depth answers comparing to the free-style unstructured prompts. GenAI should be used with the guidance of instructors.	C
13.	Neyem et al. [58]	Investigate the impact of AI-based report recommendations on the quality and effectiveness of StandUp reports in a capstone course encompassing 16 real-world projects.	Mixed (prompts' analysis + survey + topic analysis)	179 undergraduate software engineering students	AI-based recommendations can improve the overall quality of StandUp Reports. Students developed better communication skills by providing personalized feedback and recommendations for improvement.	C, B
14.	Kim et al. [71]	Explore the use of GenAI in laboratory report writing.	Qualitative (Analysis of laboratory reports + focus group)	7 undergraduate engineering students	Implementing ChatGPT in the revision writing processes improved engineering students' lab report quality. The introduction and conclusion sections were the most improved. Nonetheless, Critical thinking, reading, and composing were the least improved areas.	C, B
15.	Wilson & Nishimoto [72]	Modify programming assignments to encourage reflection of correct answers, identifying if students used ChatGPT.	Quantitative (Evaluation rubric)	160 undergraduate mechanical engineering students	In-person grading has improved communication and transparency in the courses. Through the methodology, teachers detected little evidence of ChatGPT usage by students because the emphasis was on coding understanding, error detection, and grasping new concepts.	C
16.	Knoth et al. [60]	Investigate how non-experts interact with LLM-based AI systems and engage in prompt engineering. Students were assessed in AI literacy using the guidelines proposed by Pinski & Benlian [73].	Quantitative (descriptive statistics + regression analysis)	45 undergraduate university students (15 mechanical engineering)	Prompt engineering is a GenAI quantifiable skill with implications between students who are productive in it, vs. those who are not. AI literacy and prompt engineering can enhance the learning process with the teacher's support.	C

TABLE 9. Teacher-centered proposals. Learning outcomes developed C: Cognitive, A: Affective, B: Behavioral.

#	Authors	Study aim	Research Approach	Participants	Outcomes-conclusions	Learning domains developed
1.	Sheard et al. [17]	Investigate Educators' current practices, concerns, and planned adaptations relating to GenAI tools such as ChatGPT or GitHub Copilot.	Qualitative (Semi-structured interviews, Phenomenological)	12 CS teachers	Teachers are worried about the academic integrity and challenge for the learning of novice students in programming. It is crucial to teach students how to effectively interact with GenAI tools.	C, B

TABLE 10. Table 9 continued. Teacher-centered proposals. Learning outcomes developed C: Cognitive, A: Affective, B: Behavioral.

#	Authors	Study aim	Research Approach	Participants	Outcomes-conclusions	Learning domains developed
2.	Amado-Salvatierra et al. [16]	Explore the impact of GenAI in teaching and gain understanding of the implications of GenAI in pedagogical practices.	Quantitative (Survey on Likert scale)	80 teachers. 12.5% engineering and technology teachers.	Self-assessment of educators' knowledge of AI integration indicated a moderate confidence level. Collaboration with forums and communities of practice can facilitate the sharing of successful strategies.	C, A
3, 4.	Addo [15], Addo & Sentance [55]	Investigate the motivation and how teachers are prepared to teach AI in schools. Conceptually, the study was based on Self-Determination Theory (SDT) and Self-Efficacy Theory (SET).	Qualitative (interview+ thematic analysis)	K-12 teachers (number not indicated).	The study identified the factors influencing teaching AI such as intrinsic and extrinsic drive, support community and teacher-monitored AI technologies. AI tools like ChatGPT do not support teachers' autonomy.	C, A, B
5.	Chan et al. [18]	Explore the potential role of AI-based chatbots by examining the answers of a group of teachers regarding how chatbots can improve learning in engineering education.	Qualitative (Thematic analysis)	12 software engineering teachers	GenAI Chatbots can provide students with personalized tutoring, problem-solving, and real-time support. Integrating AI-based chatbots into an engineering course has the potential to increase student engagement. Nonetheless, teachers' knowledge about the usage of GenAI is unclear.	C

middle schools' students, there is an emphasis on affective learning outcomes.

In addition, the following subsections report complementary information that we identified for these tables such as students' and teachers' perceptions regarding the GenAI tools and the learning process, alternative uses of GenAI in the methodologies, and other relevant insights.

1) STUDENTS' PERCEPTIONS

Reported positive tool-related perceptions from students are associated with the creative capability of GenAI to generate ideas and answers [35], [61], [71]; personal tutoring with the features of convenient feedback and knowledgeable assistant [63]; condensed and structured ways for programming purposes [56]; support for students with limited experience in project management [69]; debugging assistant and easy checking process for programming [62]; support for decision-making process [69]; massive, detailed, and immediate information availability [35], [58], [70]; trustworthy information comparable to experts [35]; and wiring catalyst in terms of improving professional writing style and concise and error-free language [71]. In some proposals, students have reported the GenAI features of the *help approach* comparable to human interaction [35], [62] and *not ego, not judgment* [57] referring that GenAI tools do not judge the type and level of questions that users do. These latter features of GenAI tools as conversational companions and social actors have been explained in detail in the study of Knoth et al. [60]. Even students have traits of human-to-human conversations with these tools in their prompts [60].

Positive learning process perceptions from students include the discovery of new concepts and functions through understanding the logical structure of programs [35], [56], the possibility to contrast previous knowledge with new information provided by GenAI [35], improving learning curve [61], developing analytical, problem-solving, computational, and innovative thinking [56], [69], writing catalysis by improving writing skills [71], and enjoyment and engagement in activities [58], [71].

Concerning negative tool-related perceptions, students reported hallucinations in terms of difficulty in explaining ambiguous concepts, misinterpretation, slow processing, ignoring errors in programming, misleading suggestions, or inconsistent generative responses [63]. Besides, repetitive and uncreative information [61]; limited interpretation and analysis [71]; unreliable and not trustworthy information for specific CS and engineering topics, for instance, with certain Python language libraries [35], [63], [71]. In addition, students exposed the problem of weighting the answers given by the IA because the responses provided are generic and have not been reviewed, as is the case with expert websites such as Stack Overflow [35].

Negative students' perceptions of the learning process with GenAI entail missing details and critical thinking [63], frustration in prompt generation which end up in cognitive load problems and a decrease in productivity [35], [63], [71], and instant answers against guiding the learning process that for some students can be a failure in GenAI systems [35]. Some students prefer to do the work alone rather than rely on GenAI tools [35]. Also, it has been reported that students can miss the learning purposes and goals in the courses when

they depend on GenAI usage [71]. As a result of the wrong guidance and pre-existing awareness of bias for GenAI tools, students indicated that they employed their skills to solve the problems in the courses [63] and they were not over-reliant on the GenAI tool, creating their novel and original ideas [61].

Finally, we detected some concerns and threats that the students had indicated. For instance, students have concerns about being replaced by GenAI [56], quality assurance of the information retrieved and system integration [69], and trust issues due to past inaccuracies [35], [69]. For the learning process, students manifested self-esteem and frustration problems because their task efforts were affected by rapid GenAI tool answers [56].

2) TEACHERS' PERCEPTIONS

Teachers have indicated different tool-oriented possibilities and benefits such as support for the creation of personalized programming tasks, multiple approaches to validate students' comprehension, production of large and feasible software systems, an increase of productivity, and idea generation for learning activities, content generation, productivity with time optimization, and proof reading [16]. From a pedagogical standpoint, teachers remark on the possibility that GenAI brings to rethinks and redesign the educational structure of courses through GenAI adaptation to learning goals, defining curriculum activities that incorporate GenAI, and change in traditional assessment [17]. Similarly, by using GenAI, teachers can concentrate on designing teaching activities and personalized learning experiences, promoting student's engagement [18].

Nevertheless, teachers indicate several concerns and threats. For instance, teachers manifested the misbehavior practices that students could have such as cheating and plagiarism, violation of academic integrity rules [17], missing out the focus on learning [17], over-reliance which could hinder the development of skills, e.g., in fundamental programming concepts [17], [58], lack of preparedness to cope with GenAI properly [17], and privacy, bias, and ethics issues [17].

3) METHODOLOGIES WITH DIFFERENT USES FOR GENAI

Some methodologies employed GenAI for other uses rather than only for technical purposes in computing or engineering courses. So, in K-12 levels, Garcia et al. [59] explore using GenAI with arts to critically reflect on narratives in CS and transforming spaces and shaping identities and realities of Latinx students. By the same token, Parada-Cabaleiro [67] uses GenAI to engage students as co-creators of digital media through music and storytelling. Also, Okolo [66] describes a 5-day workshop for underrepresented students, which included the understanding of ethical issues of GenAI. In these methodologies, apart from developing cognitive learning outcomes, the Authors are focused on affective ones because activities were linked with the students' identities, creativity, and cultural practices.

Aggrawal and Magana [64] employed ChatGPT to assist engineering students in conflict management and resolution through a methodology of three phases: Learn, practice, and reflect. GenAI served as a debater, generating conflict cases that students debated and solved in groups.

Shaer et al. [61] discuss the integration of GenAI in a project-based learning course for design in tangible and embodied interaction, prototyping, and co-creation. Students used DALL-E and ChatGPT during the brainstorming and rapid ideation stages.

Kim et al. [71] utilized ChatGPT to improve the writing process of engineering students in laboratory reports. GenAI could catalyze the writing process but GenAI handling should be supervised by a teacher because students can get incomplete or missing information, and they can miss the primary learning objectives and goals.

Garcia Huertes and Bragós Bardia [65] employed GenAI to analyze the CVs of graduate students and identify the possibility of entrepreneurial activity that these students could have. Offering courses related to business administration and innovation in engineering curricula could be a catalyst for future entrepreneurial activity.

4) GENAI TOOLS

Table 11 shows the GenAI tools used by the researchers and their descriptions. The list includes the authoritative tools created by the researchers. These tools have been designed as GenAI assistants to improve the teaching-learning process and benefit active learning in the courses.

5) RESEARCHERS' INSIGHTS

Several researchers uttered some insights regarding the teaching-learning process with GenAI. For educators, Knoth et al. [60] asserted that teachers must take advantage of the technological leap driven by Gen AI instead to ignoring it. Educators should have a solid understanding of how to teach students to interact with LLMs [60]. Yet, it is not clear the level of confidence and knowledge of GenAI tools by the teachers [16]. Indeed, Sheard et al. [17] put forward the lack of preparedness that teachers could have with AI. This aspect could be considered for further studies to complement professional teachers' formation [16].

In the case of students, the level of prior AI knowledge in the methodologies is not clear. In studies [35], [62], [69], [80], students counted with no experience or with a minimum level of GenAI knowledge. Conversely, in the study of Okolo [66], students were more familiar with AI technologies. This dichotomy makes it difficult to analyze if prior knowledge of GenAI impacted the learning outcomes presented, especially when they were not positive or were controversial to the extant literature (see study [63]). But certainly, what is highlighted in different studies [38], [56], [60] is the need that students to create *effective* prompts, which is seen as a new 21st-century skill.

But, at the same time, little attention has been put on how non-experts create effective prompts and how they are doing so [60]. Non-experts perceive AI as social actors and conversational companions, while higher levels of AI knowl-

TABLE 11. GenAI tools reported by the researchers in their methodologies. Ref. Reference.

GenAI Tool (Ref.)	Description
ChatGPT 3.5 and 4.0 [74]	Sibling model to InstructGPT, which is trained to follow the instructions in a prompt and provide a detailed response.
DALL-E [75]	AI system that can create realistic images and art from a description in natural language.
Promptly [56]	AI Web-based tool for visual representation of CS1 and CS2 problems. (Tool designed by the researchers).
CS50 Duck [57]	AI assistant that leverages the contextual understanding of GPT-4 to provide a truly interactive teaching and learning experience in introductory programming. (Tool designed by the researchers).
GitHub Copilot [76]	It suggests code completions as developers type and turns natural language prompts into coding suggestions based on the project's context and style conventions.
GAI-Powered Slack Bot [58]	AI-based recommendation system for StandUp software reports. (Tool designed by the researchers).
Runway ML [77]	AI tool to generate production-ready assets from complex-based simulations to hyper-realistic renders.
AIVA [78]	AI music generation assistant that allows generating songs in more than 250 styles, supporting beginners and professionals in music.
Talking Head Anime from a Single Image 3 (Tha3) [79]	Project that allows you to animate anime characters with a single image.

edge predict low rates of AI acceptance and continuance intention [60], [81]. It seems that AI knowledge is not favorable for human-AI collaboration in all environments [81]. These aspects could be a reason why in some studies mentioned, students preferred not to use or in terms of Pinski et al. [81] *delegate* tasks to Gen AI tools because they can limit their creative skills and retrieved information could be dubious, especially, for specific technical topics in programming or engineering.

C. RQ3. WHAT GEN IA INTEGRATION STRATEGIES AND ACTIVITIES HAVE EDUCATORS AND/OR RESEARCHERS CONSTRUCTED?

1) EDUCATIONAL STRATEGIES

Tables 12 and 13 describe the educational strategies elicited in the studies, most of them for computing education. In this, the typical role of GenAI tools was *assistant* for the learning process. As described, some of these strategies used their own personalized GenAI chatbots, which were tailored to the particular learning goals in the courses.

2) AI LITERACY SURVEYS, SCALES, AND AI GUIDELINES

Some AI literacy scales, surveys, and AI guidelines were identified. Table 14 shows their descriptions and references. For the AI literacy surveys, researchers show the instruments' validation with subjects and statistical methods. As for the After-Action Review for AI (AAR/AI) method, this was adapted to GenAI because it was released for AI-powered chatbots and human interaction in 2021.

3) INSTRUCTIONAL ACTIVITIES

Instructional activities for K-12 levels are characterized by participatory sessions, some of them directed by questions such as the following: How do you use AI in your daily life? What are some interesting things you could do with AI? What are some potential use cases of AI? [66]. Also, some activities reported included creating a simple ML model using the ''Teachable Machine'' platform development by Google [66]. In this activity, students gathered data of their choosing (images, sounds, poses, etc.), and trained a machine learning model. Students also learned about the ethical implications of AI. Other kinds of activities are related to examples of IA applications and the use of several resources such as fairy-tale to illustrate the learning about AI opportunities and challenges or co-creation of characters and art through GenAI tools [59], [67]. All the articles that illustrated activities comprised active learning methods, identification of AI preconceptions, participation of the students in all the sessions, and experiential learning.

Concerning instructional activities for the tertiary level, these are featured by teaching topics in programming and engineering in which GenAI tools are used typically as assistants [61], [63], [65], [70]. Instructional activities include project-based learning [69], [70], StandUp reports [58], Kata code approach [70], and feedback of students interactions leading by questions such as What do you think should happen when you use ChatGPT for this task?, What actually happened when you used ChatGPT for this task?, Why do you think ChatGPT behaved this way? [63], among others. Notably, the K-12 activities are much more descriptive in the learning strategies and activities than those for tertiary levels.

D. RQ4. WHAT ARE THE CHALLENGES, ISSUES AND OPPORTUNITIES FOR IMPROVEMENT OF GENAI IN ENGINEERING AND COMPUTING EDUCATION?

In the following, we listed several issues, challenges, and opportunities for improvement observed. We centered our attention on the pedagogical component of these apart from the technical issues indicated in other GenAI reviews and opinion studies (see references [7], [27], [53], [87]).

TABLE 12. Educational strategies identified with their description.

#	Authors	Main topic-educational level-strategy description	Primary Gen AI role (assistant, tutor, advisor, debater, etc.)	GenAI Tool
1.	Aggrawal & Magana [64]	**Strategy for conflict resolution in teamwork-University level**. The strategy included three phases: learn, practice, and reflect. In the learning phase, the students understood the conflict and how to communicate assertively their thoughts and feelings. In the second phase, students practiced conflict management using scenarios provided by ChatGPT. In the third stage, students reflect on their experiences and provide feedback related to future labor experiences.	Debater, assistant for conflict management scenarios.	ChatGPT
2.	Choudhuri et al. [63]	**Strategy for software engineering-University level**. The researchers employed the After-Action Review for AI (AAR/AI) method depicted in Table 14. The methodology included the following stages. 1. Identify the educational background of students to design AI-oriented activities aligned with learning outcomes. 2. Create program utilities to manage task complexity. 3. Disorganize code to students add comments and fix incorrect parts of it (removing code smells). 4. Deliberate inclusion of code errors.	Assistant	ChatGPT + tool created to manage task complexity.
3.	Denny et al. [56]	**Strategy for CS1 and CS2 courses in Python language-University level**. The problems only contain a visual representation thereof. Through this visual representation, students asked for a ChatGPT-based utility called *Promptly* created by the authors to guide in problem-solving.	Assistant	Promptly (Tool created by the researchers).
4.	Garcia et al. [59]	**Strategy for elementary Latinx school students-K-12 level**. The strategy involved three stages. In the first one, the researchers approached students to elicit their perception of what being Latinx means and what are the implications of technology in the community. In the second stage, students created Gen AI art stories regarding how to improve the community. In the third stage, students shared and discussed their creations.	Assistant	DALL.E
5.	Kim et al. [71]	**Strategy for writing skills in engineering courses-University level**. Students wrote their lab reports before ChatGPT incorporation. After, students rewrote their lab reports with the help of ChatGPT. A comparison was performed before and after the GenAI intervention to identify the writing skills improvement in students. Students used any prompt to interact with ChatGPT.	Assistant	ChatGPT
6.	Lauren & Watta [62]	**Strategy for SQL courses-University level**. The strategy involved well-defined learning goals and strategic assessment. Students created a database for a bookstore by completing different SQL activities, which included creating the database with a logical model. Students used ChatGPT to assist in writing the SQL schema.	Assistant	ChatGPT
7.	Liu et al. [57]	**Strategy for CS50 courses (introduction to programming)-University level**. The strategy involved the use of a personalized chatbot *CS50 duck*. The chatbot offered instant clarifications on the semantic aspects of code, as well as explained the changes for the code to the students. The chatbot was limited to the course-specific rules and guidelines.	Assistant	CS50 duck (Tool created by the researchers).
8.	Neyem et al. [58]	**Strategy for capstone software engineering courses-University level**. The strategy involved the use of a personalized chatbot that analyzed StandUp reports, providing suggestions for improvement to the students.	Assistant	AI-powered Slack bot (Tool created by the researchers).
9.	Okolo [66]	**Strategy for middle and high school students-K-12 level**. The strategy involved five sessions, viz., general introduction to AI, data and algorithms with machine learning activity, algorithm bias, introduction to generative AI, and implications of generative AI.	Assistant	ChatGPT, DALL.E.

TABLE 13. Table 12 continued. Educational strategies identified with their description.

#	Authors	Main topic-educational level-strategy description	Primary Gen AI role (assistant, tutor, advisor, debater, etc.)	GenAI Tool
10.	Ouhbi [70]	**Strategy for a maintenance programming course-University level**. The strategy worked with a Kata approach which is an exercise for programmers to develop their skills through practice and repetition. Students analyzed challenges to maintain legacy codes. Finally, students performed a project and delivered their final reports.	Assistant	ChatGPT
11.	Parada-Cabaleiro. [67]	**Strategy to promote AI-child co-creativity and AI literacy through multimedia storytelling-K-12 level**. The strategy connects children with their emotions, evaluating harmful or potential risks of AI. Children use a variety of AI tools to foster creativity skills, integrating arts and music, and reflect on its usage through guided discussions. An observation protocol is filled out by the instructors to identify the implications of AI in the educational process.	Assistant	AIVA, ChatGPT, RunwayML, Tha3.
12.	Wang et al. [38]	**Strategy for fundamentals of computer networks-University level**. The researchers proposed an isolation questioning method in which students ask ChatGPT based on the task.	Assistant	ChatGPT
13.	Wassem et al. [69]	**Strategy for software development courses-University level**. This strategy focuses on three publicly announced software projects assigned to a group. ChatGPT was utilized to support students throughout the software development process, spanning from understanding project requirements to deployment.	Assistant	ChatGPT

1) CHALLENGES AND ISSUES

- There is an *overestimation* of students' and teachers' perceptions regarding Gen AI tools but an *underestimation* of how Gen AI tools influence the teaching-learning process. We observe that many studies are focused on perceptions, but few report how GenAI tools benefit students' learning process, and how these tools are aligned with the skills and outcomes expected in the curricula of engineering and computing.
- There is no balance between the number of student-centered and teacher-centered proposals. It is missing the standpoint and criteria of teachers as co-creators of Gen AI methodologies. In fact, we did not identify educational programs focused on teachers' formation to effectively cope with Gen AI.
- There is an important amount of proposals from the U.S. vs. the rest of the world. Without other standpoints, it is tough to construct a big picture about the current practices in learning settings in different sociocultural contexts with GenAI.
- The studies focus on the affordances and technical features of the GenAI tools and do not properly consider how, based on the evidence, these tools could be a catalyst for learning, affective and behavioral skills, or a potential risk for students and teachers based on the evidence.
- Many studies lack information about students' and teachers' backgrounds regarding Gen AI tools and their prior usage. Therefore, it is tough to identify if this background influenced the learning outcomes achieved in the proposals.
- AI literacy competencies are ambiguous. For instance, some AI literacy competencies proposed by Long and Magerko [28] can work with the tertiary level (university) and not with K-12 ones, e.g., when we explain LLMs from a technical perspective. Then, it is unclear what learning outcomes and skills are expected to be developed by students with Gen AI at different educational levels.
- Learning outcomes leveraging GenAI features are focused on the cognitive domain. Few proposals at the K-12 level explore the effective and behavioral learning domains, which are also important for students to develop.
- We identified that although students can fall into the category of ''digital natives'', many of them face problems with creating proper prompts for their learning purposes.

2) OPPORTUNITIES FOR IMPROVEMENT

- We need to reflect on learning by doing during the whole process of using Gen AI. Many proposals only identified the students' perceptions once the process finished and not before. The observation concerning students' interaction should be throughout the educational process to measure the value of GenAI in creating meaningful learning. In this, positive and negative outcomes derived from GenAI usage should be analyzed.
- A better conceptualization of AI literacy and its competencies should be provided. Yet, this term and what it means for engineering and computing education is in an early development stage. A proper conceptualization

TABLE 14. AI literacy scales, surveys, and AI guidelines identified in the studies. Ref. Reference.

Scale, survey, or guideline (Ref.)	Description
AI guidelines proposed by Pinski, Adam & Benlian [81]	Guidelines based on the central concept (AI knowledge) for human-AI collaboration. The guideline emphasizes AI knowledge and edges over AI perceptions to generate replicable outcomes.
AI Literacy scale proposed by Pinski & Benlian [73]	AI literacy scale based on the dimensions of AI technology knowledge, Human actors in AI knowledge, AI steps knowledge, AI usage experience and AI design experience.
AI literacy survey proposed by Laupichler, Aster, & Raupach [82]	An item set for the assessment of non-AI experts based on 53 subject matter experts. The final version of the instrument included 38 instruments.
AI literacy survey proposed by Knoth et al. [60]	AI literacy survey based on the mentioned AI literacy scale proposed by Pinski & Benlian.
After-Action Review for AI (AAR/AI) [83]	Method to assess reinforcement learning agents in a sequential decision-making environment. The method can help to organize thoughts and think logically when people delegate decision-making to AI-based agents.
Integrative complexity score to evaluate LLM outputs [60]	A method to evaluate complexity in LLMs outputs based on the study of scaffolding and problem complexity in technology-mediated learning contexts by Janson et al. [84].
Microsoft's design guidelines for Human-AI Interaction (HAI) [85]	Eighteen guidelines encompassing different stages of interaction with AI-based systems. The guidelines explore actions when the AI system is wrong and over time.
GenderMag Method [86]	The method helps software developers & usability professionals find and fix software features with such gender-inclusiveness "bugs".

for engineering and computing should be created, analyzing the sociocultural context and the implications of GenAI in the curricula. A segmentation of the concept by educational levels, e.g., for K-12 students or undergraduate students should be made. Exploration of new scales and guidelines for AI literacy should be addressed. Comparably, it is needed to tackle what tasks in the educational process should be delegated for Gen AI processing in engineering and computing.

- Teachers' voices must be considered in the whole process of educational co-creation with GenAI. Here, the teachers' formation must be crucial for leveraging and properly integrating GenAI tools in the teaching-learning process. We distinguish between the formation in Gen AI usage and the pedagogical formation to deploy it effectively in the learning settings.
- Research on cognitive styles when students interact with AI-based systems is needed. For instance, Choudhuri et al. [63] assert that AI tools are not inclusive for cognitive styles and gender, and AI systems' approaches can differ between genders. This deserves more research and exploration and opens the discussion about the identity and sociocultural contexts [59], which matter when students use AI systems.
- We are calling for improving robustness in studies. Studies are focused on perceptions, but we need to understand the educational implications of Gen AI in engineering and computing from pedagogical insights based on the evidence.

V. LIMITATIONS

In this scoping review, we tried to perform a rigorous process as possible through the stages presented. Nonetheless, we have some limitations to expose. The first one regards the databases and search strings. We employed the keywords engineering and computing education in our search. The latter term is more accepted from K-12 levels than computer science which is often used in tertiary education. We also performed the search process in the databases mentioned. We do not discard the fact that some relevant studies have been excluded from the process, e.g., in gray literature or other languages. The second one is related to potential bias we could have when analyzing the studies. This is a proper issue of ScRs because each researcher in this study is stood up from their context and formation. We analyzed the information through a critical pedagogical lens because this was our interest in the review, using software tools to be objective in the process. However, we do not discard that in this approach, we have discarded some relevant articles. Finally, at the moment to finish this review, new articles could emerge in the databases, and unfortunately, these have been excluded. We encourage other researchers to perform new studies (empirical or reviews) that can complement the findings of this ScR.

VI. CONCLUSION AND FUTURE DIRECTIONS

This scoping review analyzed and described GenAI methodologies and strategies for engineering and computing education. We identified twenty-four proposals, most of them for computing education at the tertiary level with few for K-12 levels. At the same time, we detected that the central part of the proposals is describing students' and teachers' perceptions and affordances of GenAI tools dominated by ChatGPT. Only a few studies describe methodologies with evidence-based outcomes from quantitative, qualitative, or mixed approaches. Then, it is tough to analyze how GenAI intervenes in the learning process from cognitive, affective, or behavioral domains, including sociocultural and identity aspects of students.

Another aspect that we consider crucial to expose is that teachers' voices are missing out. Most of the proposals are student-centered and have diverse outcomes. Still, the standpoints of the teachers and formation programs for them

in GenAI at the technical and pedagogical levels are missing. Several studies asserted this issue as significant to the successful leverage of GenAI in learning settings. It seems that in the discourse, GenAI tools by themselves were self-described and self-explanatory for students when evidence shows just the opposite: Students can struggle with AI-based systems and prompt engineering, even when considered digital natives. Teacher support is essential to unveil the implications of GenAI in engineering and computing education.

Similarly, extant literature remarks on two concepts: AI literacy and prompt engineering. On one hand, seemingly, AI literacy and prompt engineering do not have a close relationship or correlation (see Knoth et al. study [60]). This indicates that students can learn how to make prompts without even knowing about AI literacy. But as Pinski et al. [81] state, high levels of AI knowledge can diminish AI-based system collaboration and delegation. This is contradictory because just the focus of AI literacy is on students comprehending AI-based systems in their technical, ethical, social, or even political dimensions. On the other hand, AI literacy drifts apart from curricula and competencies in engineering and computing, especially in tertiary education. We identified few studies describing the prior background of students with GenAI, which can suppose that the development of AI skills or AI literacy skills is delegated to other non-technical courses in the curricula. However, LLM outputs in the studies are used essentially for technical purposes at the tertiary level. Then, we advocate that a better articulation between curricula and AI literacy competencies is needed. In this regard, we could ask ourselves: What learning outcomes and skills in our courses are expected to be developed by the students, tailoring GenAI systems? Are GenAI systems compatible with these? What is our role in this?

The previous aspects point out that is necessary to take a critical pedagogical perspective of GenAI systems, and we need to think beyond the GenAI hype. Yet, several critical factors must be disclosed and researched for a better comprehension of the scopes and limitations of GenAI. GenAI is only a *tool* that should be oriented towards our pedagogical purposes and not vice versa. We encourage educators and researchers to take the outcomes and findings of this ScR to co-create methodologies, frameworks, and critical positions about what GenAI means and entails for engineering and computing education.

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JONATHAN ÁLVAREZ ARIZA (Member, IEEE) received the Technology degree in electronics from Corporación Universitaria Minuto de Dios-UNIMINUTO, in 2008, the B.Eng. degree in electronics engineering from Universidad Central of Colombia, in 2013, and the M.Ed. degree from Colombian Institution Universidad de La Salle, in 2018. He is currently an Engineering Professor with Corporación Universitaria Minuto de Dios-UNIMINUTO, Bogotá, Colombia. Since 2008, he has been a Professor with the Programs of Technology in Electronics and Industrial Engineering, Corporación Universitaria Minuto de Dios-UNIMINUTO. He has imparted courses in automation, control systems, circuits, and embedded systems. Besides, he has participated in several research projects in engineering education, open source hardware, and remote laboratories. He has published several peer-reviewed articles in recognized conferences and journals. His research interests include open source hardware and software, engineering education, control systems, embedded systems, and technology-enhanced learning.



MILENA BENITEZ RESTREPO received the master's degree in education from Universidad de Los Andes, Bogotá, Colombia, in 2011, and the Ph.D. degree in education sciences from the Pontifical Catholic University of Chile, Santiago de Chile, Chile, in 2023. She is currently a Professor of psychology with Universidad del Valle, Valle del Cauca, Colombia. She was a Psychologist with Pontificia Universidad Javeriana, Cali, Colombia, in 2003. She has also worked as a Teacher in bachelor's programs in research methodology, educational psychology, and psychological evaluation courses at the master's level course in Curriculum and Pedagogy. She has experience in educational consulting in both schools and universities. Just as she has supervised professional practices in psychology, she has directed research projects in psychology and education and has published articles in the aforementioned areas of interest. Her research interests include higher education systems, academic cultures, teaching-learning processes supported by new technologies, curricular reforms, and teacher training in higher education.



CAROLA HERNÁNDEZ HERNÁNDEZ received the degree in physics, the master's degree in education, and the master's degree in physics from Universidad de Los Andes, Bogotá, Colombia, in 2001, 2004, and 2009, respectively, and the Ph.D. degree in science and technology from Aalborg University, Denmark, in 2013. She is part of the Education and Evaluation in the Disciplines research group classified A1 in the Colombian research system. She is currently an Associate Professor with the Faculty of Engineering and Engineering Education Support Unit Coordinator, Universidad de Los Andes. She has taught courses implementing pedagogical innovations in engineering and transversal competencies, such as ethics and teamwork. Besides, she has participated in several research projects in engineering education, and she has published several peer-reviewed articles in recognized conferences and journals. Her research interests include problem-based learning oriented by projects, engineering education, teacher training, and education research.